

ARKAPRAVO GHOSH

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EDUCATION

B.Tech in Computer Science and Engineering (IoT, Cybersecurity, Blockchain)	2026
University of Engineering and Management, Kolkata	GPA: 7.5
PCMC	2022
Kendriya Vidyalaya No. 1 Saltlake	71.2%

EXPERIENCE

Full Stack Developer	May 2026 – Jun 2026
Meow AI, LLC (Ryze AI)	Remote
- Built independent CI/CD pipelines for frontend and backend services with 7+ automated test cases per pipeline, improving release reliability and reducing manual deployment validation. - Planned CPU, memory, and infrastructure requirements with the engineering team, translating application workloads into resource requirements for reliable production deployments. - Integrated Composio with the application's MCP server, enabling connections to Instagram, HubSpot, and additional third-party services through a unified integration layer. - Onboarded 3 high-value clients end-to-end and resolved production and integration tickets on demand, supporting successful adoption and maintaining service availability.	
DevOps Engineer	Nov 2025 – Mar 2026
Wiyse	Freelance, Remote
- Delivered features across 5+ production modules using React/Next.js, TypeScript, FastAPI, and SQLAlchemy, collaborating with product and engineering teams throughout the development lifecycle. - Designed and deployed autoscaling microservices on AWS EKS using ArgoCD and ECR, enabling zero-downtime deployments and automated service recovery. - Automated deployment workflows and infrastructure operations across Kubernetes workloads, reducing operational overhead and improving consistency across production releases. - Diagnosed production issues across application, container, and cloud infrastructure layers, resolving failures through systematic debugging, monitoring, and configuration changes.	
Junior Software Engineer	Aug 2025 – Sep 2025
Goodfit	Freelance, Remote
- Shipped 6 roadmap features representing 45% of Q3 scope, increasing user engagement 15% through React/Next.js modules built with TypeScript, Tailwind CSS, and PostgreSQL. - Reduced median page latency 18% by profiling application bottlenecks, refactoring rendering and data-access paths, and optimizing database interactions. - Reduced production defects per release 25% through automated testing, structured peer reviews, and tighter validation across feature development. - Collaborated across product and engineering to translate requirements into production-ready features while balancing delivery speed, maintainability, and system reliability.	
Software Development Engineer	May 2025 – Aug 2025
PredictGrowth	Freelance, Remote
- Led 70% of a performance optimization sprint and developed a multi-theme PDF export service, increasing customization usage 15% and reducing workflow latency 10%. - Increased backend throughput 35% through asynchronous workers, database query batching, caching, and workload-oriented service design. - Reduced API response times 20% by profiling bottlenecks and optimizing database access, application execution paths, and AWS service configurations. - Improved service resilience by introducing asynchronous processing and fault-tolerant AWS configurations, reducing the impact of long-running and resource-intensive workloads.	
Software Development Engineer	Feb 2024 – May 2025
RevMaxx	Part-time, Hybrid
- Designed and implemented core backend services for authentication, billing, and onboarding, supporting 120% user growth and enterprise onboarding across the platform. - Implemented role-based access control across 100% of protected endpoints, strengthening authorization boundaries for user and enterprise workflows. - Improved API and real-time streaming performance 30% through database indexing, query optimization, and WebSocket message batching. - Reduced server CPU utilization 22% by refactoring database queries, batching real-time messages, and optimizing backend processing paths.	

- Delivered 20+ full-stack features spanning analytics dashboards, API gateways, and real-time communication systems using Angular, React, Node.js, and FastAPI.
- Designed and integrated REST APIs and real-time communication channels across multiple client applications, supporting production web workloads and evolving product requirements.
- Built Docker and Jenkins CI/CD pipelines across client projects, reducing deployment time 50% through automated build, test, and deployment workflows.
- Increased critical-service test coverage to 95% by introducing automated testing into CI/CD workflows and integrating validation into the deployment lifecycle.

PUBLICATIONS

- TotalScribe: AI-Powered Clinical Documentation2026
- Developed an end-to-end open-source clinical documentation pipeline combining Whisper Large, LangChain, Mistral-7B, and FAISS RAG, reducing physician note processing latency to under 10 seconds.
 - Achieved an 8.2% Word Error Rate (WER), 0.89 BLEU score, and 0.91 F1 score for automated speech recognition, clinical SOAP note structure extraction, and diagnostic code retrieval.

PROJECTS

- Splitty:** Full-stack expense sharing application built with Next.js, TypeScript, PostgreSQL, and Drizzle ORM; features JWT authentication, automated database migrations, and real-time expense calculations.
- TotalScribe:** AI-powered medical SOAP note generator using OpenAI Whisper Large and Mistral-7B; built a full-stack Next.js + FastAPI application with SSE-based real-time updates and secure user session management.
- SATYAMEBA — HPC Cluster:** Orchestrated the SATYAMEBA high-performance computing cluster at UEM Kolkata, an academic supercomputer supporting AI research and distributed LLM workloads.
- TrainTrack API:** Built a lightweight FastAPI service that retrieves live Indian Railways status from NTES and exposes normalized, structured JSON responses for downstream applications.

SKILLS

- Programming:** JavaScript, TypeScript, Python, Bash/Shell
- Frontend:** React.js, Next.js, Angular, React Native, SCSS, Tailwind CSS, Responsive Design
- Backend & Systems:** Node.js, Express.js, FastAPI, REST APIs, WebSocket, SSE, PostgreSQL, MySQL, MongoDB, MariaDB
- Cloud & DevOps:** Linux, Docker, Kubernetes, AWS EKS, ECR, ArgoCD, Jenkins, CI/CD, Ansible, Git, GitHub, GCP, Azure, DigitalOcean
- Infrastructure & IoT:** Distributed Systems, Microservices, Raspberry Pi, Arduino, Node-RED